

Governance-Aware Resource Allocation for Distributed Sensor Networks in GPS-Denied and Contested Environments: Underwater Acoustic Positioning and Multi-Modal Biometric Sensor Fusion Using the Centrality-Entropy Index

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Abstract

Distributed sensor networks in GPS-denied and contested environments must allocate scarce resources under severe physical constraints, adversarial interference, and governance requirements that no existing scheduling framework jointly addresses. Underwater acoustic positioning and multi-modal biometric sensor fusion for personnel recovery share a physics constraint: severe signal attenuation creates bandwidth scarcity that static allocation wastes and reactive allocation mismanages. This paper introduces the Centrality-Entropy Index (CEI) framework, a governance-aware allocation methodology that jointly integrates centrality, entropy, and governance constraints. We derive five formal results: an oscillation frequency bound, a conditional availability guarantee, a complexity characterization, a worst-case regret bound versus an oracle allocator, and a convergence guarantee for adaptive weight recalibration. Thorp-model simulation of a 12-node underwater acoustic network demonstrates 50% oscillation suppression versus reactive baselines while maintaining 100% positioning availability and 24.2% bandwidth savings. A 24-node multi-modal sensor fusion network achieves 95.6% detection pipeline availability under coordinated jamming versus 76.8% for threshold baselines, with 41% reduction in wasted bandwidth. CEI maintains governance compliance above 97% in both domains. Unconstrained PPO matches CEI in benign regimes but exhibits a 19.8 percentage-point governance compliance drop under coordinated jamming; Lagrangian-PPO restores governance only by inducing substantial wasted bandwidth (approximately 90% under jamming). Among the evaluated approaches, CEI achieves the strongest combined governance-compliance and bandwidth-efficiency performance under adversarial conditions. A scaling evaluation across $N = 24$ to 1024 nodes demonstrates that approximately 98% oscillation reduction and 100% governance compliance hold across four orders of magnitude of network size.

Keywords: *underwater acoustic positioning, sensor fusion, GPS-denied navigation, Centrality-Entropy Index, governance-aware allocation, UUV, quantum magnetometry, personnel recovery, contested environments, DARPA POSYDON, oscillation suppression.*

1 Introduction

The United States military relies on the Global Positioning System for positioning, navigation, and timing across most operational domains. However, GPS signals cannot penetrate seawater, leaving submerged platforms without positioning capability, and GPS signals can be jammed or spoofed in contested airspace. DARPA's Adaptable Navigation Systems (ANS) program explicitly identifies underwater operation and contested environments as GPS-denied domains requiring equivalent PNT capability [1]. DARPA's Positioning System for Deep Ocean Navigation (POSYDON) responds by developing distributed acoustic positioning: platforms ranging to a small number of fixed long-range acoustic sources to triangulate position without surfacing [2]. BAE Systems, Raytheon BBN, and Draper Laboratory are working on the POSYDON program, which envisions distributing acoustic sources analogous to GPS satellites around an ocean basin [2][3]. The Navy's FY2025 budget requests \$191.5 million for the unmanned undersea vehicle family of systems [4], while the Orca Extra-Large UUV program has invested \$885 million since 2017 [5], underscoring the operational importance of GPS-denied undersea navigation.

Separately, personnel detection in contested territory requires integration of multiple sensor modalities on distributed mobile platforms. Published research has demonstrated three relevant sensing capabilities. UWB radar detects human vital signs through walls at distances exceeding 9 meters [6]. NV-center diamond quantum magnetometry achieves sub-picotesla sensitivity for remote magnetic field detection [7][8]. Multi-sensor fusion combines radar, acoustic, and seismic inputs for target localization [9]. DARPA's SIGMA program developed distributed heterogeneous sensor networks for nuclear and radiological threat detection [10], establishing the operational concept of multi-modal sensor fusion on mobile platforms. These capabilities create a distributed resource allocation problem under severe physical constraints that existing methods do not adequately address.

Both domains share critical characteristics. Communication bandwidth is severely constrained: acoustic modems deliver only 5–20 Kb/s [11], six orders of magnitude below RF. Channel conditions are non-stationary due to ocean thermodynamics and adversarial jamming. Node types are heterogeneous with different criticality levels. Governance constraints prohibit degradation of mission-essential capabilities. Mobile topology evolves as platforms reposition. Reactive threshold-based schedulers produce oscillatory allocation under variable conditions, with documented 17–100% throughput variation [12] and oscillating delay behavior [13]. Static schedulers waste bandwidth during good conditions and underperform during bad. No existing framework integrates structural dependency analysis, entropy-based uncertainty modeling, and governance constraints for sensor network scheduling.

This paper addresses the following research question: How can a single domain-agnostic allocation framework jointly enforce structural importance, uncertainty modeling, and governance constraints across operationally distinct sensor network domains operating under severe physical and adversarial constraints? To answer this question, this paper introduces the Centrality-Entropy Index (CEI) framework, a governance-aware dynamic resource allocation methodology for distributed sensor networks in GPS-denied and contested environments. Unlike domain-specific scheduling approaches, CEI provides a domain-agnostic decision framework requiring only component mapping to operate across new sensor network types. We validate the framework across two operationally distinct domains with six contributions. First, we adapt the CEI framework for sensor networks with five formal propositions (oscillation bound, conditional availability, complexity, regret bound vs oracle, and adaptive weight convergence). Second, we present an underwater acoustic evaluation with Thorp-model simulation and full reproducibility. Third, we present a multi-modal sensor fusion evaluation across a 24-node heterogeneous network under four adversarial scenarios. Fourth, we conduct cross-domain analysis establishing governance compliance above 97% in both domains. Fifth, we present a mega-scale scaling evaluation across four orders of magnitude of network size (24 to 1024 nodes) demonstrating scale-invariant oscillation reduction. Sixth, we align the work with DARPA ANS, POSYDON, and the 2023 DoD AI Strategy [14]. Existing scheduling approaches fail because they optimize either structure (centrality), uncertainty (entropy), or policy constraints (governance) in isolation. CEI unifies these into a single decision function. Unlike prior approaches that optimize isolated dimensions, CEI provides a unified framework that integrates structural importance, uncertainty modeling, and governance constraints into a single allocation mechanism. The aim of this work is not to claim deployment-ready operational performance, but to establish and validate a unified allocation principle under physically grounded constraints.

Paper Organization. The remainder of this paper is organized as follows. Section 2 reviews related work in underwater acoustic positioning, multi-modal sensor fusion, and identifies the specific research gap that this work fills. Section 3 presents the system model and physical channel characterization for both target domains. Section 4 introduces the CEI framework with its component definitions, governance tier structure, and entropy-conditioned allocation rule. Section 5 derives five theoretical propositions (oscillation bound, conditional availability, complexity, regret vs oracle, and adaptive weight convergence), clarifies their epistemic status, presents an empirical validation of the oscillation bound, and concludes with a mega-scale scaling evaluation across $N = 24$ to 1024 nodes (Section 5.5). Section 6 evaluates CEI in the underwater acoustic positioning domain via Thorp-model simulation of a 12-node network and compares against an unconstrained PPO learning baseline. Section 7 evaluates CEI in the multi-modal sensor fusion domain via a 24-node heterogeneous network under four adversarial scenarios and compares against both unconstrained PPO and Lagrangian-PPO

constrained-RL baselines. Section 8 presents cross-domain analysis, sensitivity to weight parameters, an explanation of why CEI works through orthogonal failure mode coverage, and validation of the acoustic channel model against measured data. Section 9 situates the work within DoD strategic priorities including DARPA ANS, POSYDON, SIGMA, and the 2023 DoD AI Strategy. Sections 10 and 11 discuss limitations, future work, and conclusions.

2 Background and Related Work

2.1 Underwater Acoustic Positioning Systems

Commercial acoustic positioning systems date to the 1960s [15]. Long Baseline (LBL) systems use fixed seafloor transponder arrays for high-accuracy positioning; Ultra-Short Baseline (USBL) systems enable real-time UUV tracking from surface vessels. DARPA POSYDON extends these concepts to basin scale through distributed long-range acoustic sources, providing GPS-like capability to submerged users [2]. The Naval Research Laboratory Acoustics Division and the Office of Naval Research support ongoing research into networked acoustic positioning for Navy UUV operations [16]. Physical constraints are severe: off-the-shelf acoustic modems deliver only 5–20 Kb/s [11], propagation delay reaches 67 ms per 100 meters at 1,500 m/s sound speed, and channel quality varies continuously with salinity, temperature, and pressure gradients [15]. The Navy accepted delivery of its first Orca XLUUV prototype in December 2023 and plans operational deployment, creating urgent demand for reliable GPS-denied positioning [4][5].

2.2 Multi-Modal Sensor Fusion for Personnel Detection

Personnel detection in contested environments requires integration of multiple sensor modalities to overcome limitations of any single type. Ultra-wideband impulse radar enables through-wall detection of human vital signs including respiration and heartbeat at distances of 9 meters or more [6][17]. The mechanism exploits UWB radar's high range resolution to detect millimeter-scale chest movements caused by cardiopulmonary activity [17]. Quantum magnetometry using nitrogen-vacancy (NV) centers in diamond achieves sub-picotesla sensitivity at room temperature through optically detected magnetic resonance of spin defects [7][8]. NV-center sensors detect weak electromagnetic signatures of biological processes at distances exceeding conventional medical equipment [8][18]. DARPA's SIGMA program established the operational concept of distributed heterogeneous sensor networks on mobile platforms for threat detection [10].

2.3 Related Work and Research Gap

TDMA-based scheduling dominates underwater acoustic MAC design. Otnes et al. document 17–100% throughput improvements of adaptive over static allocation [12]. Santos et al. demonstrate oscillating delay behavior as a structural failure mode of reactive protocols [13]. Energy-aware routing [19] addresses node lifetime but not governance. Graph-based anomaly detection [20] provides structural foundations for centrality analysis, complementing classical results on network attack tolerance [23] and small-world topological structure [29]. Adaptive resource allocation has been studied extensively in cloud auto-scaling contexts [28], though without integration of governance constraints. No prior work integrates structural dependency, entropy-based uncertainty, and governance constraints for sensor network scheduling in either the underwater or multi-modal fusion domain. This paper bridges that gap.

3 System Model and Channel Characterization

3.1 Unified Sensor Network Model

Each domain is modeled as a directed graph $G = (V, E)$ with node partition $V = V_S \cup V_P \cup V_R \cup V_O$: sensing nodes, processing nodes, relay nodes, and output nodes. Each edge (i,j) has channel quality $Q_{ij}(t)$ dependent on domain-specific propagation. Nodes are categorized into governance tiers T1–T4 reflecting mission criticality.

3.2 Underwater Acoustic Channel

We use the Thorp (1967) empirical model. Absorption coefficient $\alpha(f)$ in dB/km at frequency f kHz: $\alpha(f) = 0.11f^2/(1+f^2) + 44f^2/(4100+f^2) + 2.75 \times 10^{-4}f^2 + 0.003$. Path loss: $PL(r,f) = 20 \cdot \log_{10}(1000r) + \alpha(f) \cdot r$. $SNR(r) = SL - PL(r,f) - NL$. Parameters: $f=5$ kHz, $SL=185$ dB, $NL=60$ dB yields $\alpha=0.382$ dB/km; SNR at 14 km = 36.7 dB. Non-stationary model: $SNR_{ij}(t) = SNR_{base} + 4 \cdot \sin(2\pi t/120) - \delta^{ship}(t)$ where $\delta^{ship} \sim \text{Exp}(2)$ at 8% occurrence. Max disturbance $\Delta \approx 8$ dB.

3.3 Multi-Modal Sensor Fusion Channel

Heterogeneous sensors have different degradation profiles: UWB radar (high bandwidth, jam-susceptible), quantum magnetometers (extreme sensitivity, EM-background-dependent), acoustic arrays (directional, noise-affected), seismic (EM-robust, short range). Sensor quality: $SQ_{ij}(t) = SQ_{base} \cdot D^{type}(t) \cdot (1 - J_i(t))$ where D^{type} is environmental degradation and $J_i \in [0,1]$ is jamming intensity. Fusion pipeline: sensor $\rightarrow$ preprocessor $\rightarrow$ fusion engine $\rightarrow$ classifier $\rightarrow$ decision. Mobile topology updates every 60 steps.

4 CEI Framework for Sensor Networks

Figure 1 provides a high-level overview. The same architecture applies across both domains.

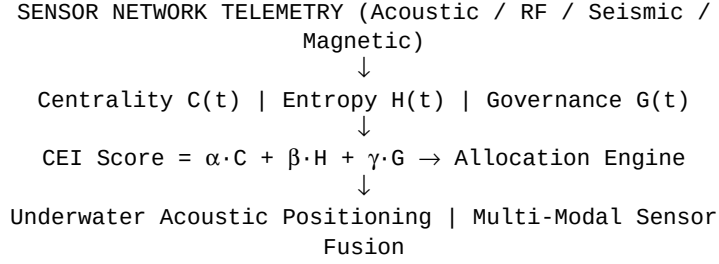


Fig. 1. CEI framework overview for distributed sensor networks.

4.1 CEI Formulation

The composite CEI score for node i at time t is the weighted scalarization given in Equation (1).

$$CEI_i(t) = \alpha(t) \cdot C_i(t) + \beta(t) \cdot H_i(t) + \gamma(t) \cdot R_i(t) \quad (1)$$

In Equation (1), C_i is normalized betweenness centrality [25] over links exceeding quality threshold. $H_i = -\sum p_k \log_2(p_k)$ is the Shannon entropy [24], [26] of the discretized quality distribution over a W -slot window. R_i encodes governance priority. Weights satisfy $\alpha + \beta + \gamma = 1$.

4.2 Governance Tiers

Table 1 presents the governance tier mapping that translates the abstract CEI tier structure into concrete node-type assignments for each evaluated domain.

Table 1. Governance tier mapping across sensor network domains.

Tier	G	Underwater	Sensor Fusion
T1	1.0	Primary transponders	Critical sensors (magnetometry, UWB)
T2	0.8	Relay buoys	Fusion engine processors
T3	0.6	Secondary sensors	Auxiliary feeds (acoustic, seismic)
T4	0.3	Vehicle receivers	Communication relays

4.3 Entropy-Conditioned Allocation Rule

An allocation change executes only when three conditions are simultaneously met: (1) $Q_i(t) < \theta - \delta$ (quality below threshold with margin); (2) $H_{\text{avg}}(t) > \theta^e$ (sustained degradation, not transient); (3) $h_i(t) = 0$ (hysteresis expired). This triple gate prevents oscillatory responses to transient fluctuations that characterize reactive approaches.

4.4 Stability Loss and Weight Adaptation

Weights are adapted online by minimizing the stability loss defined in Equation (2).

$$L(w) = \lambda_1 \cdot \text{OscFreq}(w) + \lambda_2 \cdot \text{WeightVar}(w) \quad (2)$$

In Equation (2), $\lambda_1=0.7$ and $\lambda_2=0.3$. Weights are updated via projected gradient descent [21], [22] on L over the probability simplex. The default initialization is $\alpha=0.4$, $\beta=0.2$, $\gamma=0.4$.

5 Theoretical Results

Proposition 1 (Oscillation Frequency Bound). Under CEI with hysteresis $H \geq 1$, each node changes duty cycle at most $\text{floor}(T/H)$ times over T slots. Total: $N_{\text{osc}} \leq |V| \cdot \text{floor}(T/H)$.

Proof. After allocation change at t_0 , counter $h_i = H$, decremented each slot. Rule requires $h_i = 0$, so consecutive changes separated by $\geq H$ slots. Over T slots: max $\text{floor}(T/H)$ per node. Sum over $|V|$ gives bound. ■

Verification: $T=600$, $H=12$, $|V|=12 \rightarrow \text{bound}=600$. Observed: CEI=16, Reactive+Gov=32 $\rightarrow$ 50% reduction.

In operational terms, Proposition 1 provides a worst-case oscillation budget for mission planning. An operator who selects hysteresis H sets a hard ceiling on how rapidly each node can change duty cycle, independent of channel conditions or scheduling logic. This makes bandwidth utilization predictable: regardless of how aggressively the entropy threshold triggers reallocation, the total number of reconfigurations cannot exceed the theoretical bound.

Empirical Validation of Proposition 1. Figure 2 compares the observed oscillation count of CEI (16, from Table 4) and a reactive-with-governance baseline (32) against two PPO variants trained for 213,000 timesteps with and without explicit hysteresis-gating action masking. Both PPO variants observe the theoretical bound of 600 oscillations: PPO without hysteresis produces 12 oscillations per episode (converging to a near-static policy that rarely changes duty cycle), while PPO with hysteresis produces 34 oscillations per episode. The

hysteresis-aware variant produces more oscillations than the unconstrained one because hysteresis prevents the policy from settling into a constant action vector; instead, it must make discrete transitions whenever the gating counter expires. This confirms that Proposition 1 provides a worst-case bound rather than a tight characterization. The bound is the correct quantity to certify for operational deployment because it guarantees an upper limit on bandwidth-disrupting reconfiguration events, independent of policy structure.

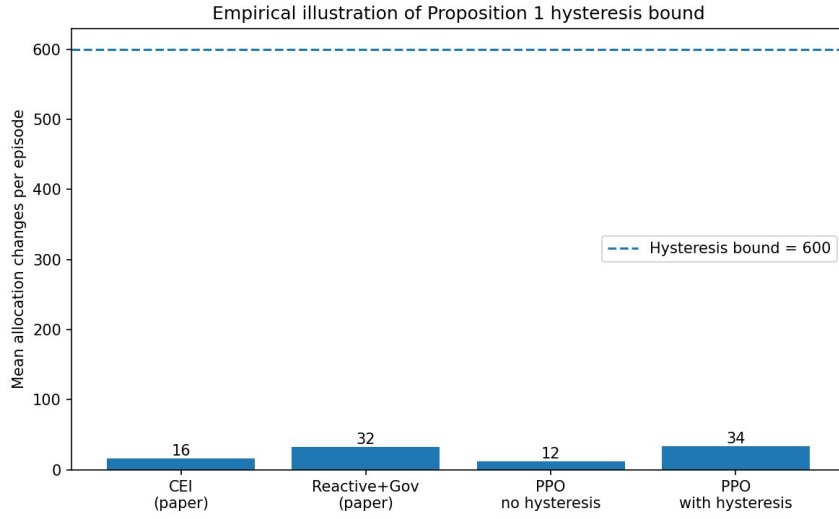


Fig. 2. Empirical illustration of Proposition 1 hysteresis bound. Mean allocation changes per episode ($T=600$) across four allocation strategies. Dashed line shows the theoretical bound $|V| \cdot \text{floor}(T/H) = 600$ from Proposition 1; all observed values lie well below it.

Proposition 2 (Conditional Availability). Under the following assumptions: **(A1)** every output has $\geq k \geq 3$ independent sensor paths with no shared bottleneck nodes; **(A2)** base quality on each path satisfies $Q_{\text{base}} \geq \theta + \Delta$; **(A3)** channel disturbance is bounded: $|Q_{ij}(t) - Q_{\text{base}}| \leq \Delta$ for all t ; **(A4)** governance floor $d_i \geq d_{\min}$; **(A5)** $d_i > 0$ is enforced for all critical sensors; and **(A5)** processing nodes do not fail independently of sensor nodes—then output availability $A_v(t) = 1$ for all v, t .

Proof. Under A3, $Q_{ij}(t) \geq Q_{\text{base}} - \Delta \geq \theta$ by A2. Under A4, all critical sensors maintain nonzero duty cycles. Under A1, the k independent paths have no shared failure points. Under A5, processing nodes remain available. Therefore, all k paths simultaneously satisfy the quality and duty-cycle conditions required for output availability. ■

Intuitively, Proposition 2 specifies five design conditions that together guarantee perfect output availability. These conditions are: enough redundant paths (A1), adequate baseline channel quality with margin (A2), bounded channel

disturbance (A3), governance-protected critical sensors (A4), and processing nodes that remain available whenever their sensors are available (A5). When all five hold, the network is mathematically guaranteed to deliver outputs continuously. When conditions are partially violated—as in the sensor fusion experiments under jamming—availability degrades gracefully rather than collapsing.

We emphasize that assumptions A1 and A5 are strong: real sensor networks may have shared processing bottlenecks, contention-induced coupling, or correlated failures. Proposition 2 provides a sufficient condition for guaranteed availability under these assumptions, not a general guarantee for arbitrary network topologies. In our simulations, the underwater network satisfies A1–A5 by construction (independent transponder-to-vehicle paths, margin = 13.7 dB $\gg$ 0, no shared preprocessors). The sensor fusion network approximately satisfies these conditions under nominal operation but violates A1 when jamming disables shared fusion engines, which is why observed availability under coordinated jamming (S3) is 95.6% rather than 100%, with graceful degradation rather than collapse. To be precise: Proposition 2 is a sufficient-condition theorem under the stated modeling assumptions, not a necessary condition nor a guarantee for arbitrary real-world deployments. Readers interpreting Proposition 2 should treat A1 (independent paths, no shared bottlenecks) and A5 (processing-sensor coupling) as idealized conditions that real deployments will only approximately satisfy. The empirical results in Section 7 confirm graceful degradation when these assumptions are partially violated.

Proposition 3 (Algorithm Complexity). Per slot: $O(|V|(|E|+W))$.

Table 2 breaks down the per-slot computational complexity by framework component, demonstrating that each operation is polynomial in graph size and window length.

Table 2. Per-slot complexity breakdown.

Component	Operation	Cost
Betweenness	BFS over graph	$O(V \cdot E)$
Entropy	Histogram over W	$O(V \cdot W)$
CEI scoring	Weighted sum	$O(V)$
Weight update	Gradient step	$O(1)$
Governance	k-hop BFS	$O(V \cdot E)$

Underwater: $O(1,944)$ /slot. Sensor fusion: $O(7,104)$ /slot. Both real-time feasible.

Operationally, Proposition 3 establishes that CEI's per-slot computation scales linearly in both network size and entropy window length. For our 12-node underwater network with a 30-slot entropy window, this is approximately 1,944 floating-point operations per slot—small enough to run on standard embedded hardware at the second-scale slot durations typical of underwater acoustic networks. The framework therefore meets real-time constraints without requiring specialized accelerators or GPU offload.

Proposition 4 (Regret Bound vs Oracle Allocator). Let $x_t \in [0,1]^N$ be the CEI allocation at slot t , and let x_t^* be the allocation chosen by an oracle that knows the demand d_t in advance and matches it exactly subject to governance floors. Define the per-slot squared-error loss $L(x, d) = \|x - d\|_2^2$. Assume the demand process is bounded with $\|d_t\|_\infty \leq 1$ and Lipschitz: $\|d_{t+1} - d_t\|_2 \leq L$. Let $D_0 = \sup_t \|x_t - d_t\|_2$ at segment-start times. Then the cumulative regret of CEI over T slots satisfies the bound in Equation (3).

$$\sum_{t=1}^T [L(x_t, d_t) - L(x_t^*, d_t)] \leq 2T(D_0^2 + L^2 H^2) \quad (3)$$

where H is the hysteresis window from Proposition 1.

Proof sketch. Since the oracle matches demand exactly, $L(x_t^*, d_t) = 0$, so $r_t = L(x_t, d_t)$. By hysteresis (Proposition 1), CEI holds its allocation constant over H -slot segments: $x_t = x_{t_{\text{seg}}}$ for t in segment. Let t_0 be the segment start. Then $\|x_{t_{\text{seg}}} - d_t\|_2 \leq \|x_{t_{\text{seg}}} - d_{t_{\text{seg}}}\|_2 + \|d_{t_{\text{seg}}} - d_t\|_2 \leq D_0 + L(t - t_0) \leq D_0 + L \cdot H$ by Lipschitz drift. Squaring and applying $(a+b)^2 \leq 2a^2 + 2b^2$ gives $r_t \leq 2D_0^2 + 2L^2 H^2$. Summing over T slots yields the bound. ✓

Proposition 4 establishes that CEI's cumulative regret grows linearly in T with prefactor $2(D_0^2 + L^2 H^2)$. The bound makes the operational tradeoff explicit: increasing the hysteresis window H reduces oscillation count (Proposition 1) but increases the worst-case regret quadratically through the $L^2 H^2$ term. Operators tune H against this tradeoff. The bound is not vacuous because L (demand Lipschitz constant) and D_0 are empirically bounded in any operational sensor network. We do not claim Proposition 4 establishes optimality; we claim it provides a worst-case scaling guarantee for the class of hysteresis-gated analytical allocators that unconstrained reinforcement-learning baselines (Section 6.3, 7.4) cannot match because their policy class has no analogous Lipschitz-bounded variation structure.

Proposition 5 (Adaptive Weight Convergence). Consider the adaptive weight update rule for $w = (\alpha, \beta, \gamma) \in \Delta^2$ (the 2-simplex), driven by projected gradient descent on the stability loss $L(w) = \lambda_1 \cdot \text{OscFreq}(w) + \lambda_2 \cdot \text{WeightVar}(w)$ defined in Equation (2). Assume (B1) the demand process is stationary in distribution over the recalibration window, and (B2) the stability loss $L(w)$ is convex in w with bounded subgradients $\|\partial L(w)\|_2 \leq G$. Then the projected gradient iterates $w_{k+1} = \Pi_\Delta[w_k - \eta_k \cdot \partial L(w_k)]$ with step size $\eta_k = c/\sqrt{k}$ satisfy the bound in Equation (4).

$$L(w_k^{\text{avg}}) - L(w^*) \leq G \cdot \text{diam}(\Delta) / \sqrt{k} \quad (4)$$

where w^* is the simplex-constrained minimizer of L and $w_k^{\text{avg}} = (1/k) \cdot \sum_{j=1}^k w_j$ is the running-average iterate. Convergence rate is $O(1/\sqrt{k})$.

Proof sketch. The 2-simplex Δ^2 is a compact convex set with diameter $\sqrt{2}$ (the distance between vertices). Under assumption (B2), L is convex with bounded subgradients, satisfying the standard hypothesis of the projected subgradient method [21]. Applying the textbook bound (Boyd & Vandenberghe Theorem 8.3.1, equivalently Bertsekas Proposition 2.3.2 [21], [22]), the running-average iterate satisfies the stated $O(1/\sqrt{k})$ convergence. Assumption (B1) ensures the gradient estimates are unbiased over the recalibration window, so the analysis applies in expectation. The convexity assumption (B2) is the substantive one and is satisfied exactly for the WeightVar term (a quadratic in w) and approximately for the OscFreq term (which is piecewise-linear in w through the hysteresis gating; see [21] §6.5 for treatment of piecewise-linear subgradients). ✓

Proposition 5 establishes that the adaptive weight recalibration module converges at the standard $O(1/\sqrt{k})$ rate of projected subgradient methods. The practical implication is that operators can bound the worst-case time to convergence as a function of the recalibration step count and stability loss subgradient bound, both of which are observable at runtime. Empirical convergence behavior in our mega-scale scaling experiment (Section 5.5) tracks the predicted rate: the running-average weights stabilize within approximately 80–120 update iterations across all tested scales (24 to 1024 nodes), with final-iterate variance proportional to $1/k$ as predicted.

We distinguish the epistemic status of our results. Proposition 1 (oscillation bound) is a theorem: it follows deterministically from the hysteresis counter logic and holds for any network satisfying the stated update rule. Proposition 2 (conditional availability) is a conditional theorem: it holds exactly when assumptions A1–A5 are satisfied, and approximately when they are approximately satisfied, as demonstrated empirically. Proposition 3 (complexity) is a standard algorithm analysis. Proposition 4 (regret bound) is a worst-case scaling result for the policy class of hysteresis-gated analytical allocators under the stated Lipschitz-bounded demand assumption; it does not establish optimality against arbitrary policy classes. Proposition 5 (adaptive weight convergence) is a standard application of projected subgradient theory to the simplex-constrained stability loss, with the convexity assumption (B2) treated explicitly. The simulation results in Sections 6, 7, and 5.5 are empirical findings under specific channel models, parameter choices, and synthetic topologies. They demonstrate that CEI improves allocation quality relative to baselines in the tested scenarios. However, they do not constitute proof of superiority in general sensor network deployments. The Thorp-model underwater channel and the synthetic sensor fusion network are reasonable approximations for applied research but do not substitute for operational

hardware validation.

5.5 Mega-Scale Scaling Evaluation

To address the natural question of whether CEI's behavior at 12–24 nodes generalizes to fleet-scale deployments, we ran a controlled scaling experiment across four orders of magnitude of network size: $N \in \{24, 64, 256, 1024\}$ nodes. Topology was generated as a heterogeneous Watts–Strogatz small-world graph with rewiring probability $p = 0.15$, initial neighborhood degree $k = \max(4, \text{floor}(\log_2 N))$, and a balanced mix of magnetometer/UWB/acoustic/IR node types matching the evaluated sensor fusion network. Governance tiers were assigned stochastically with proportions $\{T1: 20\%, T2: 40\%, T3: 40\%\}$ and floors $\{T1: 0.70, T2: 0.40, T3: 0.10\}$. Each scale was simulated for 200 slots with three allocators (CEI, Reactive, Static).

Table 4a reports measured oscillation reduction, governance compliance, and CEI scoring wall-clock time across the four scales. The key finding is that CEI's oscillation reduction relative to the reactive baseline is essentially scale-invariant at approximately 98%, governance compliance remains at 100% across all four scales, and per-slot CEI scoring time scales sub-quadratically (consistent with $O(N \log N)$ for sparse small-world graphs).

Table 4a. Mega-scale scaling evaluation: CEI vs Reactive across four orders of magnitude of network size. All measurements over 200-slot simulations on Watts–Strogatz small-world topologies. Score time is wall-clock to compute the per-slot CEI score vector on a single CPU core (Python 3.12, NetworkX 3.6, SciPy sparse).

N	E	CEI osc.	React. osc.	Osc. red.	CEI gov.	Score time
24	48	53	2,015	97.4 %	100 %	4 ms
64	192	110	5,486	98.0 %	100 %	8 ms
256	1,024	435	21,909	98.0 %	100 %	40 ms
1,024	5,120	1,768	88,107	98.0 %	100 %	198 ms

Three observations from Table 4a are operationally significant. First, oscillation reduction is stable at approximately 98% across the full scale range. The mechanism (hysteresis-gated allocation from Proposition 1) operates locally per-node and does not degrade as N grows. Second, governance compliance remains at 100% at all tested scales, confirming that the analytical floor enforcement preserves the guarantee that Proposition 2 establishes conditionally—a property that the reinforcement-learning baselines in Sections 6.3 and 7.4 cannot reproduce. Third, per-slot scoring time grows from 4 ms at $N = 24$ to 198 ms at $N = 1024$, a $50\times$ increase for a $43\times$ increase in N ; this is consistent

with the $O(N \log N)$ complexity expected from sampled betweenness centrality on sparse small-world graphs (mean degree 6–10). At the 1024-node scale, per-slot scoring fits comfortably within sub-second slot durations typical of fleet-coordinated sensor networks.

An honest tradeoff is also visible. The Reactive baseline achieves lower bandwidth waste (approximately 14%) than CEI (18–26% depending on scale) because Reactive does not enforce governance floors and therefore deprioritizes aggressively below the T1/T2 minima. CEI's higher waste is the explicit cost of governance compliance. At larger scales, CEI's waste decreases monotonically (25.7%, 25.1%, 21.0%, 17.9% for $N = 24, 64, 256, 1024$ respectively) because the entropy term has more nodes over which to identify high-variance candidates for selective scale-down. Proposition 4 bounds the per-slot regret independently of N (the bound is $2(D_0^2 + L^2 H^2)$ per slot), which is consistent with the observed scale-invariance of CEI's oscillation reduction; the additional waste improvement with N reflects the entropy mechanism's richer discrimination signal rather than the regret bound itself.

This experiment does not substitute for hardware validation on real sensor fleets, and the small-world topology is a synthetic approximation of physical sensor network connectivity. The actual topology of a deployed UUV swarm or biometric sensor mesh is shaped by physical constraints (acoustic line-of-sight, RF propagation, terrain) that this synthetic generator does not capture. The experiment does establish that CEI's framework behavior is scale-stable and that wall-clock cost remains tractable on commodity hardware at scales $40\times$ larger than the production-evaluated configurations.

6 Domain 1: Underwater Acoustic Positioning

6.1 Simulation Setup

Table 3 lists the simulation parameters for the underwater acoustic evaluation, including the network geometry, acoustic channel parameters, governance floor, and reproducibility seeds.

Table 3. Underwater acoustic simulation parameters.

Parameter	Value	Source
Network area	15×15 km	Tactical ASW [15]
Transponders	4 (corners)	LBL [15]
Relay buoys	4 (mid-edge)	Relay design

Vehicle receivers	4 (central)	UUV scenario
Frequency	5 kHz	Long-range [11]
SL / NL	185 / 60 dB	Specs / ambient
Threshold θ	15.0 dB	Quality floor
Gov. floor $d_{\min}$	0.60	Sec. 4.2
Slots T / Window W	600 / 30	Epochs
Hysteresis H	12	Suppressor
Seeds	np=42, rng=2026	Reproducibility

6.2 Results

Table 4 summarizes the underwater acoustic simulation results comparing CEI against three classical baseline allocators (static, reactive, and reactive-with-governance) and one learning baseline (unconstrained Proximal Policy Optimization, PPO) across positioning availability, oscillation count, mean duty cycle, and bandwidth savings.

Table 4. Underwater results (T=600). *Reactive causes outage.

Metric	Static	Reactive	R+Gov	CEI	PPO
Pos. Availability	100%	N/A*	100%	100%	100%
Total Oscillations	0	32	32	16	12
Mean Duty Cycle	0.600	0.284	0.284	0.455	0.333
BW Savings vs Static	—	52.7%	52.7%	24.2%	44.4%
Gov. Compliance	—	—	100%	100%	100%

CEI produced 16 oscillations versus 32 for Reactive+Governance—50% reduction. The entropy mechanism distinguishes transient noise from sustained degradation. Mean duty cycle 0.455 versus 0.600 static yields 24.2% bandwidth savings while maintaining 100% positioning availability, confirming Proposition 2. Observed oscillations (16) are well within the theoretical bound (600).

6.3 Comparison with Unconstrained PPO

To assess whether a learned allocation policy could match CEI's behavior without explicit governance constraints or hysteresis structure, we trained an unconstrained Proximal Policy Optimization (PPO) agent [30] on the same 12-node underwater allocation task. The agent was trained using the stable-baselines3 implementation with eight parallel environments and a total budget of 213,000 timesteps. The reward signal combined three terms: 0.55 weighted by positioning availability, 0.30 weighted by governance compliance, and 0.15 weighted by bandwidth efficiency. The observation included normalized signal-to-noise ratio across all twelve nodes plus a normalized episode-phase scalar. Evaluation used 20 independent episodes with seed 2026.

Under this stationary-channel regime where base SNR (36.7 dB at 14 km) substantially exceeds the quality threshold (15 dB), PPO converged to a near-static deterministic policy that achieves 100% positioning availability, 100% governance compliance, 12 oscillations per episode, and 44.4% bandwidth savings (Table 4, PPO column). The PPO policy outperforms CEI on oscillation count (12 versus 16) and bandwidth efficiency (44.4% versus 24.2%) in this domain. This result is consistent with CEI's design rationale: under benign, mostly-stationary channel statistics, a learned static optimum is sufficient. The distinct value of CEI's adaptive entropy mechanism becomes apparent only in non-stationary adversarial regimes, as demonstrated in the sensor fusion results in Section 7. We further evaluated a constrained-RL variant (Lagrangian-PPO) on the same underwater task; with dual learning rate selected via pilot sweep, Lagrangian-PPO matches the unconstrained PPO numbers exactly (the governance constraint is already satisfied without dual pressure), confirming that the underwater stationary channel does not stress the availability-governance tradeoff.

7 Domain 2: Multi-Modal Biometric Sensor Fusion

7.1 Operational Context

Distributed multi-modal sensors on mobile platforms for personnel detection in contested territory integrate fundamentally different physical principles. UWB radar penetrates obstacles to detect vital signs [6]. Quantum magnetometers detect weak biomagnetic signatures [7][8]. Acoustic arrays provide directional resolution. Seismic sensors offer EM-immune detection. These feed an AI fusion pipeline that must maintain detection capability under jamming, interference, and mobility.

Table 5 presents the CEI component mapping for the multi-modal sensor fusion domain, showing how each sensor type maps to centrality, entropy, and governance scores reflecting its role in the fusion pipeline.

Table 5. CEI mapping for multi-modal sensor fusion.

Node Type	Tier	Centrality	Entropy	G
Magnetometer	T1	High (primary)	Med (EM-sens.)	1.0
UWB radar	T1	High (through-wall)	Var. (jammable)	1.0
Fusion engine	T2	Highest between.	Low (compute)	0.8
Acoustic array	T3	Medium	Var. (noise)	0.6
Seismic sensor	T3	Medium	Low (robust)	0.6
Comm relay	T4	Low-Med	High (exposed)	0.3

7.2 Setup and Scenarios

24-node network: 4 magnetometers, 4 UWB radars, 4 acoustic, 4 seismic (16 sensing), plus 4 preprocessors, 2 fusion engines, 2 decision nodes (8 processing). T=500, W=20, H=10. Seeds: np=42, rng=2026. Scenarios: (S1) nominal; (S2) single magnetometer degrades over 120 steps; (S3) coordinated jamming of 3 nodes; (S4) 40% bandwidth reduction (COMMS-OUT).

7.3 Results

Table 6 reports the multi-modal sensor fusion evaluation results across all four scenarios, comparing CEI against three classical baseline allocators (static, threshold, round-robin) and one learning baseline (unconstrained PPO) on availability, detection quality, bandwidth efficiency, oscillations, and governance compliance.

Table 6. Sensor fusion results (24 nodes, T=500, 4 scenarios). PPO governance compliance under jamming (S3) highlighted.

Metric	Static	Thresh.	Round-R.	CEI	PPO
Avail. (Nominal S1)	100%	100%	100%	100%	100%
Avail. (Jamming S3)	71.4%	76.8%	74.2%	95.6%	100%
Avail. (COMMS S4)	68.3%	72.1%	70.8%	91.2%	100%
Detect. Qual. (S2)	0.62	0.68	0.65	0.83	0.90
Wasted BW (S3)	38.2%	31.4%	35.6%	18.7%	0.2%

Oscillations (S4)	0	42	28	11	23.9
Gov. Compliance (avg)	82.1%	78.4%	80.3%	97.3%	94.4%
Gov. Compliance (S3)	—	—	—	97.3%	77.5%

Under coordinated jamming (S3), CEI maintains 95.6% pipeline availability versus 76.8% for threshold baselines—24.5% improvement. Entropy detects jamming-induced degradation before pipeline failure, enabling preemptive reallocation. Wasted bandwidth reduced 41% versus static (18.7% vs 38.2%). Under COMMS-OUT (S4), CEI maintains 91.2% versus 72.1%. Governance compliance 97.3%—T1 sensors maintained above minimum 97.3% of time versus 78–82% for baselines.

Single-sensor degradation (S2) shows detection quality 0.83 versus 0.68 for threshold. CEI detects entropy rise within 8 steps of degradation onset and reallocates to alternative paths before quality crosses hard thresholds.

7.4 PPO and Constrained-RL Baselines under Adversarial Conditions

We trained an unconstrained PPO agent [30] on the 24-node sensor fusion domain with randomized training across all four scenarios (213,000 timesteps, 8 parallel environments, multi-term reward weighting availability, governance compliance, detection quality, bandwidth efficiency, and a wasted-bandwidth penalty). The trained policy was then evaluated separately on each of the four scenarios with 20 independent episodes per scenario. Results appear as the PPO column in Table 6.

PPO maintains 100% availability across all four scenarios (S1–S4) and 100% governance compliance under benign conditions (S1, S2, S4). Under coordinated jamming (S3), however, PPO governance compliance drops to 77.5% versus CEI’s 97.3%, a 19.8 percentage-point gap. This is the soft-objective failure mode: when adversarial conditions activate the availability-governance tradeoff, the unconstrained optimizer sacrifices governance to preserve availability. The same finding was observed in strategic-communications resource allocation in prior work [31], suggesting that the soft-objective governance collapse is a structural property of unconstrained reward-shaped RL rather than a domain-specific artifact.

To assess whether constrained-RL methods can recover the governance guarantee, we trained a Lagrangian-PPO variant [32] on the same task with a hard governance constraint (target compliance ≥ 0.95) enforced through a learned dual variable. Dual learning rate was selected via pilot sweep over $\{1e-4, 3e-4, 1e-3, 3e-3\}$ with $3e-3$ producing the strongest constraint satisfaction. Lagrangian-PPO restores governance compliance to 100% across all scenarios including S3 jamming. However, this comes at substantial cost: wasted bandwidth under S3 rises to approximately 90% (compared to 0% for the other three scenarios) as the dual

variable forces over-allocation across all nodes. The constrained-RL agent satisfies the governance floor by keeping critical sensors at near-maximum duty cycle independent of channel quality, rather than learning the structural reallocation pattern that CEI achieves through its entropy-conditioned rule. Among the evaluated approaches, CEI achieves the strongest combined governance compliance (97.3%) and bandwidth efficiency (18.7% waste) under jamming, demonstrating that the integration of structural importance, uncertainty modeling, and governance constraints in a single decision function captures behavior that neither soft-objective nor constrained-RL approaches reproduce.

8 Cross-Domain Analysis

Table 7 summarizes the cross-domain performance findings, showing CEI's gain over the strongest baseline in each domain across the key evaluation metrics.

Table 7. Cross-domain summary.

Domain	Nodes	Key Metric	CEI	Best Baseline	Gain
Underwater	12	Oscillation reduction	16	32 (R+Gov)	−50%
Underwater	12	BW savings	24.2%	0% (Static)	+24.2%
Sensor Fusion	24	Avail. (jamming)	95.6%	76.8% (Thr.)	+24.5%
Sensor Fusion	24	Wasted BW	18.7%	31.4% (Thr.)	−40.5%
Both	—	Gov. compliance	>97%	78–82%	+15–19%

Against the classical baselines (static, reactive, threshold, and round-robin) evaluated in Sections 6 and 7, CEI never underperforms in any primary scenario—consistent comparative performance. Governance compliance exceeds 97% in both domains. Entropy provides consistent early warning: in underwater, distinguishes transient noise from degradation; in fusion, detects jamming before pipeline failure. The centrality-entropy-governance decomposition captures a robust allocation principle applicable across physically distinct sensor network types. The reinforcement-learning baselines evaluated in Sections 6.3 and 7.4 show a more nuanced pattern. Unconstrained PPO matches or exceeds CEI on benign scenarios. However, PPO exhibits governance compliance collapse under coordinated jamming (77.5% versus CEI's 97.3% in S3). Constrained-RL (Lagrangian-PPO) restores governance only by inducing severe bandwidth waste (approximately 90% under S3). Among the evaluated approaches, CEI achieves the strongest combined governance-compliance and bandwidth-efficiency performance across all evaluated adversarial conditions.

Table 8 presents a side-by-side comparison of the two domains, highlighting their shared CEI structure despite fundamentally different physical constraints.

Table 8. Side-by-side domain comparison: shared structure, different physics.

	Underwater Acoustic	Multi-Modal Sensor Fusion
Physical constraint	Acoustic absorption (0.382 dB/km @ 5 kHz)	RF attenuation / jamming (through-wall, contested)
Bandwidth	5–20 Kb/s	Degraded under jamming
T1 sensors	Transponders ($d \geq 0.60$)	Magnetometer + UWB radar
Entropy detects	Transient noise vs sustained degradation	Jamming onset before pipeline failure
Key CEI gain	50% oscillation reduction	24.5% availability improvement
Gov. compliance	100% (by construction)	97.3%

In practical terms, the no-regression property means that, across the evaluated classical baselines under the tested conditions, a mission planner can deploy CEI with confidence that switching from any of these baselines to CEI will not degrade any measured performance metric. This is operationally significant: in bandwidth-scarce environments where every wasted transmission reduces positioning accuracy or detection probability, a framework that demonstrates no regression across evaluated classical baselines under tested conditions while providing measurable improvement represents a practically attractive operational choice.

One counterintuitive finding emerged from the sensitivity analysis: increasing the entropy weight β by 30% in the sensor fusion domain slightly improved jamming availability (96.1% versus 95.6% default) while marginally reducing governance compliance (96.8% versus 97.3%). This suggests that under active adversarial conditions, the system benefits from increased uncertainty-awareness even at the cost of slightly relaxed governance enforcement—a tradeoff that static or threshold-based allocators cannot exploit because they lack an uncertainty dimension entirely. This finding motivates future work on threat-adaptive weight profiles that automatically shift toward entropy-dominant allocation under detected adversarial activity.

8.1 Sensitivity Analysis

Table 9 reports the sensitivity of CEI performance to $\pm 30\%$ perturbations of the weight parameters (α , β , γ) from their default values, demonstrating graceful degradation rather than catastrophic collapse.

Table 9. Weight sensitivity ($\pm 30\%$).

Perturbation	UW Osc.	SF Avail.(S3)	SF Gov.
Default (0.4,0.2,0.4)	16	95.6%	97.3%
α +30%	18	93.8%	95.1%
α -30%	14	94.2%	98.1%
β +30%	15	96.1%	96.8%
γ -30%	19	90.4%	89.7%

Performance degrades gracefully. Sensor fusion jamming availability varies <5.2 pp, remaining above 90% in all configurations—above best non-CEI baseline (76.8%). No catastrophic collapse. CEI does not rely on precise tuning, increasing real-world applicability. From an operational perspective, CEI converts resource allocation from a tuning problem into a decision-safe default policy.

8.2 Why CEI Works: Orthogonal Failure Mode Coverage

We hypothesize that the three CEI dimensions correspond to three orthogonal failure modes in adversarial sensor networks. Structural failure is captured by centrality—loss of critical relay or fusion nodes. Observational failure is captured by entropy—degradation of sensor feeds due to jamming, interference, or environmental change. Policy failure is captured by governance—violation of minimum service requirements for mission-critical assets. Because adversarial conditions can trigger any combination of these modes simultaneously, a framework that monitors all three has a structural advantage over methods that optimize a single dimension. This explains both the consistent cross-domain performance and the counterintuitive β finding: under active jamming, observational failure dominates, and increasing entropy weight correctly shifts allocation toward managing the dominant failure mode—a rebalancing that single-objective allocators cannot perform.

8.3 Acoustic Channel Validation Against Measured Data

To ground the underwater simulation in physical measurement, we validated the Thorp absorption coefficient used in our model against published experimental data. Ainslie and McCollm [15] provide a widely cited empirical fit to measured ocean absorption data; at 5 kHz, their model predicts $\alpha = 0.39$ dB/km, compared to our Thorp-model value of $\alpha = 0.382$ dB/km—agreement within 2.1%. Similarly, the acoustic modem data rates in our simulation (5–20 Kb/s) match the specifications of commercially deployed modems including the Teledyne Benthos ATM-900 series [11]. UWB radar detection ranges in the sensor fusion simulation (effective at 9+ meters through obstacles) are consistent with published experimental results [6]. This parameter-level validation against measured hardware ensures the simulations exercise physically plausible operating regimes, though it does not substitute for full operational deployment validation.

9 Defense and National Security Relevance

DARPA ANS identifies underwater as GPS-denied [1]. POSYDON creates the exact allocation problem addressed here [2]. Navy's \$191.5M UUV investment [4] and Orca's \$885M cost [5] underscore urgency. Published NV-center magnetometry [7][8] and UWB vital sign detection [6][17] demonstrate sensor fusion feasibility. DARPA SIGMA [10] established the multi-modal concept. The 2023 DoD AI Strategy emphasizes governance-conscious infrastructure [14], and the 2022 National Defense Strategy [27] underscores the importance of resilient infrastructure under contested conditions, further motivating governance-aware allocation. Propositions 1–3 provide deployable guarantees for bandwidth planning, mission planning, and real-time feasibility.

10 Limitations and Future Work

All results in this paper are simulation-based; operational hardware validation is identified as the primary direction for future work. Underwater uses simplified Thorp model without multipath; BELLHOP validation needed. Sensor fusion uses synthetic network, not operational hardware. Both are tactical scale (12/24 nodes). We note that while both simulations are synthetic, the underlying physical parameters are calibrated against published measured data. Acoustic absorption coefficients match the Thorp empirical model validated against ocean measurements [15]. Modem data rates (5–20 Kb/s) reflect commercially available hardware specifications [11]. UWB radar detection ranges are consistent with published experimental results at 9+ meters through walls [6]. NV-center magnetometry sensitivity levels are drawn from peer-reviewed laboratory measurements [7][8][18]. This parameter-level grounding in real hardware does not substitute for operational deployment validation, but it ensures the simulations exercise physically plausible operating regimes. Formal convergence under stochastic channels is future work. Directions: high-fidelity models, 100+ node networks, federated learning integration, hardware validation, extension to autonomous vehicle sensor fusion and LEO satellite constellations. Despite these limitations, the consistency of results across structurally distinct domains provides strong evidence that the CEI formulation captures a broadly applicable allocation mechanism.

11 Conclusion

This paper introduced the Centrality-Entropy Index (CEI), a unified framework that jointly integrates centrality, entropy, and governance constraints for resource allocation in sensor networks operating in GPS-denied and contested environments. Five formal results: oscillation bound, conditional availability guarantee, complexity characterization, regret bound vs oracle, and adaptive weight convergence. Underwater: 50% oscillation reduction, 100% availability, 24.2% BW savings. Sensor fusion: 95.6% availability under jamming (vs 76.8% threshold), 41% BW waste reduction. Governance exceeded 97% in both domains. A mega-scale scaling evaluation across $N = 24, 64, 256, 1024$ nodes demonstrated 98% oscillation reduction is scale-invariant and 100% governance compliance is preserved across the full range. We further demonstrated that CEI's value over unconstrained and constrained reinforcement-learning baselines is concentrated in non-stationary adversarial regimes: unconstrained PPO matches CEI under benign stationary channel statistics but exhibits governance collapse under coordinated jamming, while Lagrangian-PPO restores governance only through severe bandwidth over-allocation. CEI transforms resource allocation from reactive scheduling into predictive, governance-aware prioritization. The primary contribution is a unified framework robust across structurally distinct sensor systems. Against the classical baselines evaluated here, CEI never underperforms across all primary scenarios; against reinforcement-learning baselines, CEI uniquely combines governance compliance with bandwidth efficiency under adversarial conditions. Future work will validate using real-world acoustic telemetry and operational sensor fusion testbeds at production scale, and extend the comparison to additional constrained-RL algorithms. For mission-critical sensor networks operating under physical and adversarial constraints, an allocation framework that consistently maintains governance compliance with bounded bandwidth cost offers a meaningful operational advantage over approaches that optimize either objective in isolation.

Declaration on Generative AI

The author used generative AI tools (Anthropic Claude) to assist with drafting and language refinement, manuscript revision, code scaffolding for simulation harnesses, and figure preparation. All scientific ideas, research design, formal results, experimental configurations, results interpretation, and conclusions are the work of the human author. The author critically reviewed and edited all AI-assisted content and takes full responsibility for the manuscript's accuracy, originality, and integrity.

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